

APS110 – Lecture 5

See hand written notes for drawings.

Units...

- Stress (σ) – typically for metals, strength values are in the MPa range (megaPascal)
- Young's modulus (E) – typically for metals, E values will be in the GPa range (gigaPascal)

Remember, the slope of a stress (σ) vs. strain (ϵ) curve is Young's modulus

Young's modulus vs. Density

- Density (ρ) = mass per unit volume
 - o Density [=] $\text{kg/m}^3 = \text{g/mL} = \text{g/cm}^3$ or g/cc
- E vs. ρ graph → experiment plotting all the world's materials on this chart
 - o Water has a density of 1 → good reference point (what floats/sinks in water?)
 - o Ceramics tend to have similar density to metals, but with a higher E
 - o There tends to be a trade-off between density and Young's modulus
 - We want to try and fill the void of material that has high modulus and low density → lots of research effort
 - CFRP – carbon fibre; starts to enter into this region
 - o There is a similar relationship observed in strength vs. density

Density

- Want to understand the structure of solids